

Maintenance Report: Change Log and Refactoring

ExpenseMate – Personal Expense Manager

Software Construction, Assignment 1 | Phase 5: Maintenance & Evolution | Release 1.0 → 1.1, 27 September 2026

1 new feature (adaptive)	2 bugs fixed (corrective)	6 refactorings (perfective)	144 / 144 automated tests passing
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1. Change Log

ID	Type	Change	Verified by
CH-01	Adaptive	Multi-currency. Enter amounts in PKR, USD, EUR, GBP, AED or SAR; original amount kept, PKR value used for all totals. Currency picker with “≈ Rs” preview; GET /api/currencies; additive migration.	6 server tests, client tests, browser check
CH-02	Corrective	D-12: TRUST_PROXY setting so the login limiter sees real clients behind a proxy.	Rate-limit test with two clients
CH-03	Corrective	D-13: blank lines no longer shift CSV error line numbers.	Regression test (failed first)
CH-04	Perfective	Renamed 3 files to match the naming standard.	Full suite
CH-05	Perfective	Shared <i>commonValidator</i> for month and id rules (was 3 copies).	Full suite
CH-06	Perfective	<i>monthRange</i> moved to <i>utils/dates</i> ; removes a service-to-service dependency.	Full suite
CH-07	Perfective	Removed duplicate query <i>sumExpensesByCategory</i> .	Full suite
CH-08	Perfective	Transaction form logic moved to <i>lib/transactionForm.js</i> ; <i>FieldError</i> component. Complexity 25 → 11.	8 new client tests
CH-09	Perfective	<i>useConfirm</i> hook and <i>ensureCategory/findOwnTransaction</i> helpers replace repeated code.	Full suite

After every change: 105 server and 39 client tests pass, and the production build succeeds.

2. Refactoring Report

Measure	v1.0	v1.1	Change
Highest function complexity	25	17	–32%
Transaction form complexity	25	11	–56%
Average complexity	2.03	1.99	Lower
Duplicated logic	12 copies	5	–7
Files breaking naming standard	3	0	–3
Automated tests	128	144	+16

Remaining: seven React components stay above 10 because of conditional markup; splitting them is the next step. *updateTransaction* rose from 4 to 7 due to currency conversion (Law II).

3. Lehman's Laws

Version	What changed	Code (lines)	Files	Tests
v0.1	Project start	14	2	1
v0.5	Backend API complete	429	23	24
v0.8	CSV import/export	631	25	41
v0.9	Frontend added	2,849	55	41
v1.0	Accounts and demo login	3,664	67	59
v1.1	Phases 3–5 (tests, fixes, multi-currency, refactoring)	3,753	73	144

Law	Evidence	Response
I. Continuing change	New needs each phase: accounts, per-user data, multi-currency.	Changes stay within one or two layers.
II. Increasing complexity	Code 429 → 3,753 lines; updateTransaction 4 → 7.	Measured and refactored the worst function (25 → 11).
III. Self-regulation	Tests grew with code (24 → 144); about 3.4 defects per KLOC.	Every change ships with tests.
IV. Organisational stability	Steady pace of small commits in every phase.	Work split by module.
V. Conservation of familiarity	Apart from v0.9 (frontend), each release added under 900 lines.	Small releases, one concern per commit.
VI. Continuing growth	2 → 73 files; v1.1 goes beyond the SRS.	Features added as new modules.
VII. Declining quality	Login change broke phone navigation (D-11).	Regression and system tests before release.
VIII. Feedback system	Phase 4 metrics chose this phase's work.	Measure → change → re-measure.